

Lab 14: Fully Bayesian Models

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14.1 A Bayesian NFL Power-Rating Model

Use regular-season games from `14_nfl-games.csv` to fit a single-season version of the power-rating model from lecture:

$$y_i \sim \mathcal{N}(\beta_0 + \beta_{H(i)} - \beta_{A(i)}, \sigma_{\text{game}}^2),$$

$$\beta_j \sim \mathcal{N}(0, \sigma_{\text{team}}^2).$$

The starter Stan file centers team strengths so that β_0 is interpretable as league-wide home-field advantage.

In this lab, each posterior draw is one plausible football world after seeing the 2023 regular season: one home-field advantage, one value of game randomness, one value of team-strength spread, and one strength for every team.

14.1.1 Task 1: Fit and Diagnose

Filter to the 2023 regular season, construct team indices, and fit the model with at least four chains. Report:

- warnings from Stan;
- $\hat{R}$ and effective sample size for key parameters;
- the number of divergent transitions;
- trace plots for β_0 , σ_{game} , and σ_{team} .

Do not interpret the model until the diagnostics are satisfactory.

14.1.2 Task 2: A Posterior Distribution Over the Leaderboard

Plot posterior means and 95% credible intervals for every team's strength. Then use posterior draws to calculate:

1. $\mathbb{P}(\beta_{\text{DAL}} > \beta_{\text{PHI}} \mid \text{data})$;
2. each team's probability of being top five;
3. the team with the highest posterior mean and the team most likely to rank first.

Explain why the last two answers need not match.

14.1.3 Task 3: Posterior Prediction

Choose a neutral-site matchup. For every posterior draw, simulate a future score differential and estimate:

- expected score differential;
- predictive win probability;
- a 95% posterior predictive interval.

Compare the predictive interval with the credible interval for the difference in team strengths.

Suppose a sportsbook makes Team A a neutral-site favorite by s points, so Team A covers when its future score differential is greater than s . Find the largest value of s for which the posterior predictive probability that Team A covers is at least 55%. Explain how this decision differs from asking whether Team A is stronger.

14.1.4 Task 4: Posterior Predictive Checks

Generate replicated score differentials from the fitted model. Compare replicated and observed:

1. score-differential distributions;
2. home win rates;
3. rates of games decided by at least 21 points.

Identify one systematic feature the model misses.

Name two omitted football variables that could matter for prediction and explain how each could enter an extended model.

14.1.5 Task 5: Prior Sensitivity and Extension

Change the prior on σ_{team} to be substantially tighter or wider. Refit and explain how the leaderboard changes. In particular, explain how this prior changes pooling toward league average.

Finally, describe how you would extend the model to all seasons using the dynamic prior

$$\beta_{j,s} \sim \mathcal{N}(\gamma\beta_{j,s-1}, \sigma_{\text{season}}^2).$$